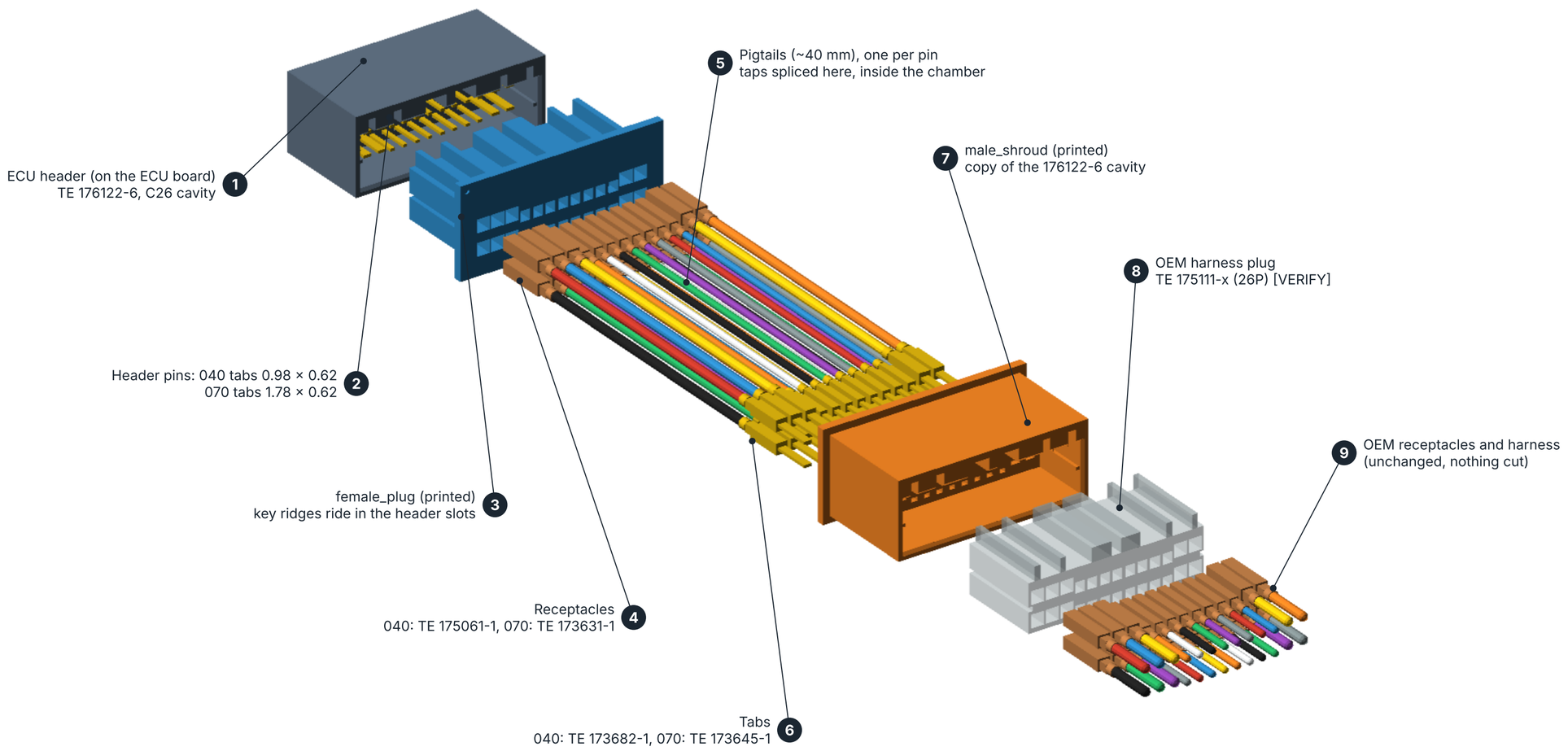
1KZ-TE ECU interposer: C26 section, exploded view and cross-section analysis

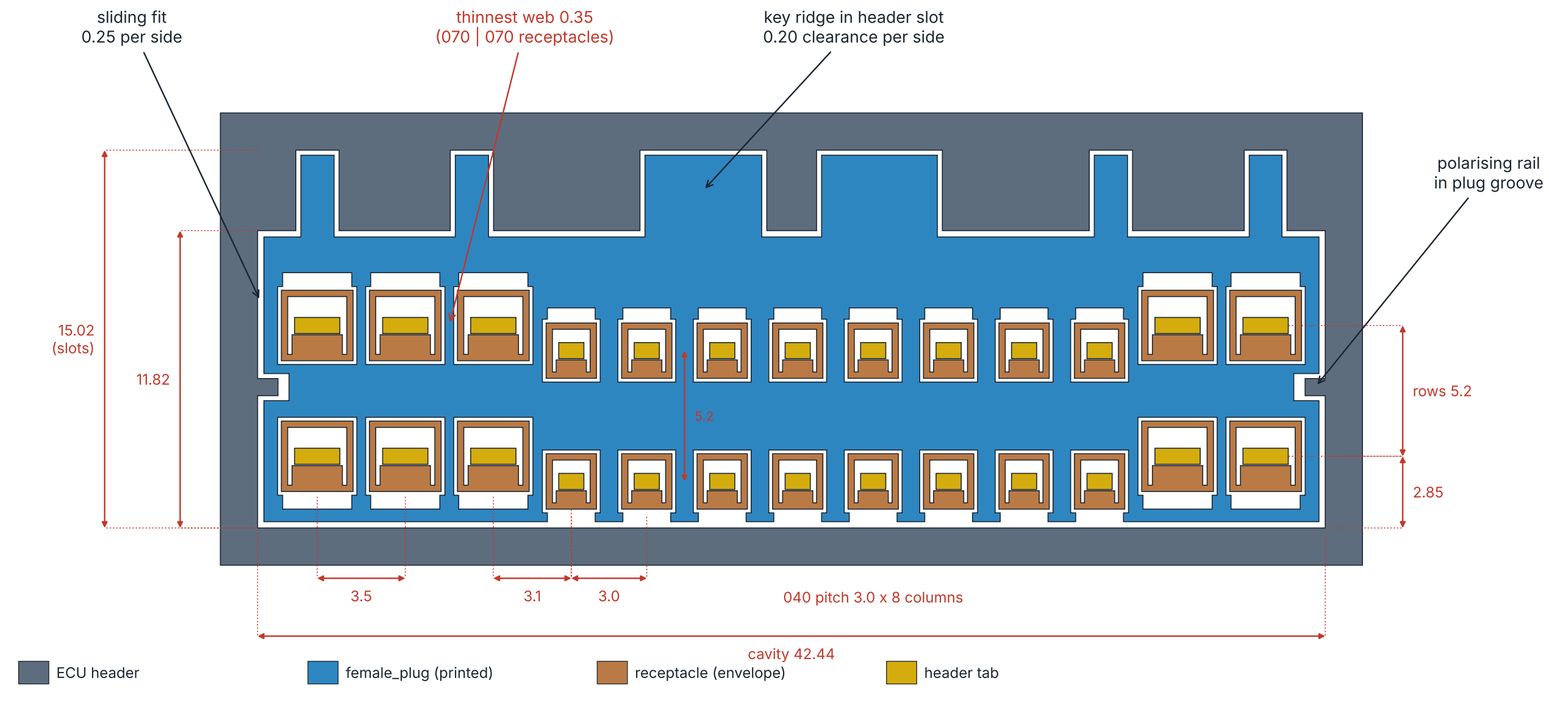
Header TE 176122-6 (geometry measured from TE's 3D model). Terminal bodies and the harness latch are estimates [MEASURE]. All dimensions in mm.

1 Exploded view (mating axis left to right)



The two-piece chamber that clamps both flanges is hidden here; it is shown in section 3.

2 Section A-A: interposer plug seated in the ECU header, cut 4 mm above the header floor



Analysis

Fit in the header

Plug envelope vs cavity	41.94 × 11.32 in 42.44 × 11.82 (0.25/side)
Key ridges vs slots	0.20/side, unique pattern per section
Insertion depth	14.3 of 14.5 (stops 0.2 short of the floor)

Pin engagement

040 tab into receptacle	5.89 (tab 7.09 proud - gap 0.2 - face wall 1.0)
070 tab into receptacle	6.79 (tab 7.99 proud)
Shroud side	identical by construction (same tab heights)

Walls in the printed plug

Between columns	070 | 070: 0.35 070 | 040: 0.37 040 | 040: 0.70
Between rows	040: 2.70 070: 2.10
Under lower row (incl. lance)	040: -0.10 070: +0.50
Over upper row (incl. lance)	040: 2.82 070: 1.42

Verdict

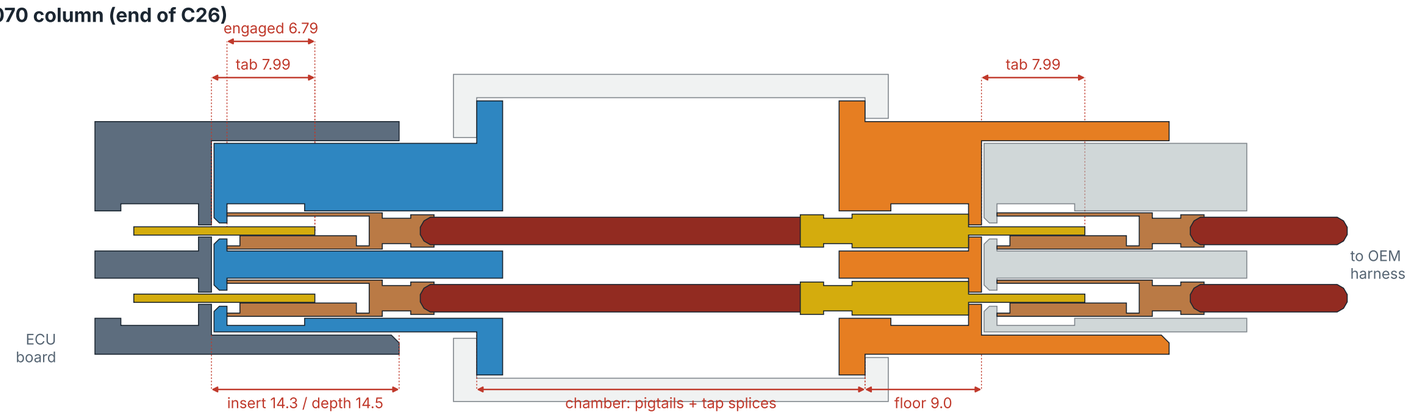
Fits and engages as designed. Walls under 0.6 mm need resin or MJF.

Lower 040 lance reliefs break 0.1 mm through the underside: they become open windows, which is harmless but visible.

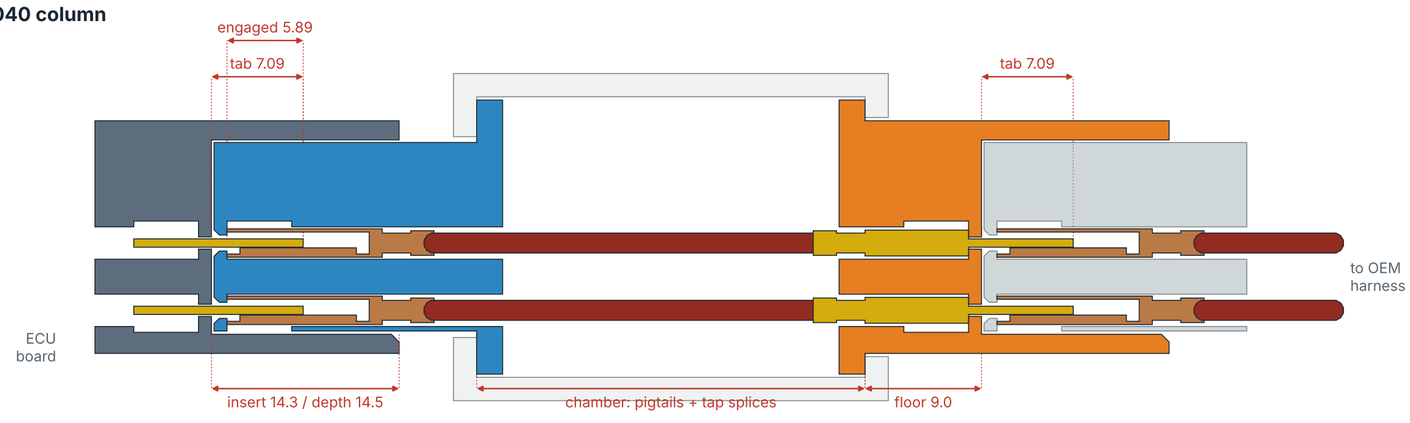
Not checked yet: latch retention and terminal body sizes [MEASURE].

3 Section B-B: lengthwise through one pin column, assembled (ECU on the left, harness on the right)

070 column (end of C26)



040 column



Generated by hardware/ecu_interposer/diagram/make_diagram.py from cad/diagram.scad. Sections are true scale.